Biology practice paper: answers and marks

How to mark your paper. Go through one question at a time. Read the answer, then check your own answer against "How the marks are given". Give yourself the mark if your answer means the same thing, even if you used different words.

Write your mark next to each question, then add them up. Read the "Watch out" boxes even when you got the marks: they're the mistakes that cost people marks in the real exam.

01.1

1 mark

Which structure is found in a plant cell but not in an animal cell? Tick one box.

The answer

Chloroplast.

How the marks are given

- 1 mark: you ticked chloroplast.

Watch out

- Only tick one box. If you tick two, you get no mark.
- Mitochondria and ribosomes are in animal cells too, and every cell has a cell membrane.

01.2

2 marks

A student looks at a cheek cell through a light microscope. The image of the cell is 6 mm wide. The real cell is 0.06 mm wide. Calculate the magnification. Show your working.

The answer

×100. Working: magnification = image size ÷ real size = 6 ÷ 0.06 = 100.

How the marks are given

- 2 marks: the answer 100 (or ×100). You get both marks for the right answer even if you didn't show working.
- 1 mark: if your answer is wrong, but you wrote down 6 ÷ 0.06.

Watch out

- Divide the image size by the real size, not the other way round. 0.06 ÷ 6 gives 0.01, which is wrong.
- Both sizes must be in the same units before you divide. Here they're both in mm.

01.3

2 marks

Before a cell divides by mitosis, it goes through a stage of growth. Describe two things that happen to the cell during this stage.

The answer

The cell grows bigger and copies its DNA (its chromosomes). It also makes more ribosomes and mitochondria.

How the marks are given

- 1 mark for each of these, up to 2 marks:
- the cell grows (gets bigger)
- the DNA or chromosomes are copied
- more sub-cellular structures are made, such as ribosomes or mitochondria

Watch out

- "The nucleus splits" doesn't count. That's mitosis itself, not the stage before it.

02.1

2 marks

One cylinder had a mass of 2.50 g at the start and 2.80 g at the end. Calculate the percentage change in mass. Show your working.

The answer

+12%. Working: change in mass = 2.80 - 2.50 = 0.30 g. Percentage change = 0.30 ÷ 2.50 × 100 = 12%.

How the marks are given

- 2 marks: the answer 12 (or +12). You get both marks for the right answer even with no working.
- 1 mark: if your answer is wrong, but you worked out the change (0.30) and divided by the starting mass.

Watch out

- Divide by the starting mass (2.50), not the end mass. 0.30 ÷ 2.80 gives 10.7, which is wrong.
- The mass went up, so the answer is +12, not -12.

02.2

1 mark

Between which two concentrations in the table would a cylinder show no change in mass?

The answer

Between 0.2 and 0.4 mol/dm³.

How the marks are given

- 1 mark: both numbers, 0.2 and 0.4.

Watch out

- Look for where the results change from + (gained mass) to - (lost mass). That's where there's no change.

02.3

3 marks

Explain why the cylinder in the 0.8 mol/dm³ solution lost mass.

The answer

The sugar solution was more concentrated than the inside of the potato cells. So water moved out of the cells by osmosis, from the dilute solution inside the cells to the more concentrated solution outside, through the partially permeable cell membranes.

How the marks are given

- 1 mark: you say water moved out of the potato by osmosis.
- 1 mark: you say the solution was more concentrated than the inside of the potato cells (or the cells were more dilute).
- 1 mark: you say water moves from a dilute solution to a more concentrated one through a partially permeable membrane.

Watch out

- It's water that moves, not sugar. "The potato lost sugar" or "sugar moved in" gets no marks.

02.4

1 mark

Why did the student calculate the percentage change in mass, rather than just the change in mass?

The answer

Because the cylinders started with different masses. Percentage change lets you compare them fairly.

How the marks are given

- 1 mark: you say the starting masses were different, or that it lets you compare the results fairly.

Watch out

- "It's more accurate" on its own isn't enough. Say why.

03.1

2 marks

Name the enzyme that breaks down lipids, and name the products.

The answer

Lipase breaks lipids down into fatty acids and glycerol.

How the marks are given

- 1 mark: lipase.
- 1 mark: fatty acids and glycerol. You need both for this mark.

Watch out

- Don't mix it up with protease (proteins into amino acids) or amylase (starch into sugars).

03.2

2 marks

Explain why an enzyme stops working at a high temperature.

The answer

At a high temperature the enzyme's active site changes shape (the enzyme is denatured), so the substrate no longer fits into it.

How the marks are given

- 1 mark: you say the active site changes shape, or the enzyme is denatured.
- 1 mark: you say the substrate no longer fits the active site.

Watch out

- Never say the enzyme is "killed" or "dies". Enzymes aren't alive, and that gets no marks.

03.3

2 marks

Name the reagent used to test for protein, and give the colour it turns if protein is present.

The answer

Biuret reagent. It turns from blue to purple if protein is there.

How the marks are given

- 1 mark: Biuret.
- 1 mark: purple (lilac or mauve also count).

Watch out

- Iodine is the test for starch and Benedict's is the test for sugars. Saying "blue" doesn't count: that's the colour before the test.

04.1

2 marks

Explain how a stent helps a person with coronary heart disease.

The answer

A stent is a small mesh tube put into a narrowed coronary artery. It holds the artery open, so blood carrying oxygen can get through to the heart muscle.

How the marks are given

- 1 mark: you say it keeps the coronary artery open (or widens it).
- 1 mark: you say this lets blood (and oxygen) reach the heart muscle.

Watch out

- A stent goes in an artery, not a vein.
- A stent doesn't lower cholesterol. That's what statins do.

05.1

1 mark

Measles is caused by a virus. Explain why a doctor would not give antibiotics to treat measles.

The answer

Antibiotics only kill bacteria. They don't work on viruses, because viruses live and reproduce inside your cells.

How the marks are given

- 1 mark: you say antibiotics kill bacteria but not viruses.

Watch out

- "The virus is resistant" is wrong. Antibiotics never work on any virus.

05.2

3 marks

Explain how a vaccination can stop a person becoming ill with measles.

The answer

The vaccine contains a small amount of dead or inactive measles virus. Your white blood cells respond by making antibodies against it. If the real measles virus gets in later, your white blood cells recognise it and make the right antibodies quickly, so the virus is destroyed before you become ill.

How the marks are given

- 1 mark: you say a dead or inactive form of the virus is put into the body.
- 1 mark: you say white blood cells make antibodies.
- 1 mark: you say that next time the antibodies are made quickly, so the virus is destroyed before you're ill.

Watch out

- The vaccine doesn't contain antibodies. It makes your body produce them.

06.1

6 marks

A student wants to find out how light intensity affects the rate of photosynthesis in pondweed. Describe a method the student could use. Include how they would make it a fair test.

A full-marks answer

Put some pondweed in a beaker of water. Put a lamp 10 cm away and measure the distance with a ruler. Leave it for 5 minutes to settle. Count the bubbles of oxygen the pondweed gives off in one minute. Then move the lamp to 20 cm, 30 cm, 40 cm and 50 cm and count the bubbles for one minute at each distance. Repeat each distance three times and work out the mean. To make it a fair test, keep the temperature the same (use an LED bulb, which doesn't get hot), use the same piece of pondweed, and keep everything else the same apart from the distance of the lamp.

How the marks are given

This question is marked as a whole. Read your answer, then pick the description that fits it best.

5 to 6 marks	Your method is clear, in the right order, and would work. You say how you change the light (moving the lamp), how you measure the rate (counting bubbles or measuring gas in a set time), at least two things you keep the same, and that you repeat it.
3 to 4 marks	Your method would mostly work, but a step is missing or unclear. You change the light and measure something, and keep at least one thing the same.
1 to 2 marks	You give some steps, but not in a clear order, or you don't say how you measure the rate.
0 marks	Nothing that would help find the answer.

Check you included

- Pondweed in water
- Lamp at a set distance, measured with a ruler
- Count bubbles (or measure the gas) in a set time
- Change the distance and do it again
- Repeat and work out a mean
- Keep the same: temperature, pondweed, time for each count, no other light

Total: 30 marks. A rough guide for this practice paper: 22 or more is a strong result. 15 to 21 is solid, with some gaps to fix. Under 15 means going back over these topics in the notebook first. Whatever you got, write down the questions where you lost marks: those are what to revise next.